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College of Osteopathic Medicine

Pathology: Neoplasia of the Kidney

PPOM2 (Principles and Practices of Osteopathic Medicine 2)



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Session Objectives

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By the conclusion of the session, the student will be able to:

1. Describe and discuss the etiologies, pathogenesis, pathology, and findings **in benign renal tumors** including **Adenoma, Angiomyolipoma, and Oncocytoma**
2. Describe and discuss the etiology, pathogenesis, pathology, and findings in **Renal Cell Carcinoma**
3. Describe and discuss the etiology, pathogenesis, pathology, and findings in **Wilms tumor**
4. Describe and discuss the etiology, pathogenesis, and findings in **Transitional Carcinoma of the renal pelvis**

Source: Course Syllabus

Kidney Tumors

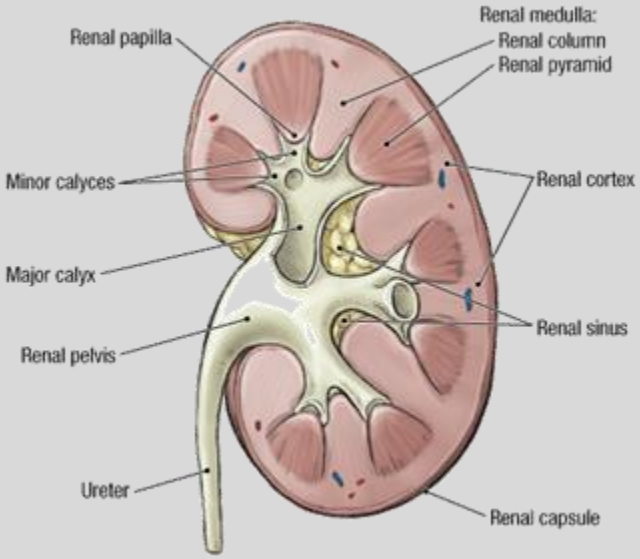
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Benign

- Renal (Papillary) Adenoma
- Angiomyolipoma
- Oncocytoma

Malignant

- Renal Cell Carcinoma
- Urothelial (Transitional Cell) Carcinoma of Renal Pelvis
- Wilms Tumor (mainly pediatric)



Robbins & Cotran Pathologic Basis of Disease, 10th edition-
Chapter 20: The Kidney; Grants Dissector, 16e

Adenoma

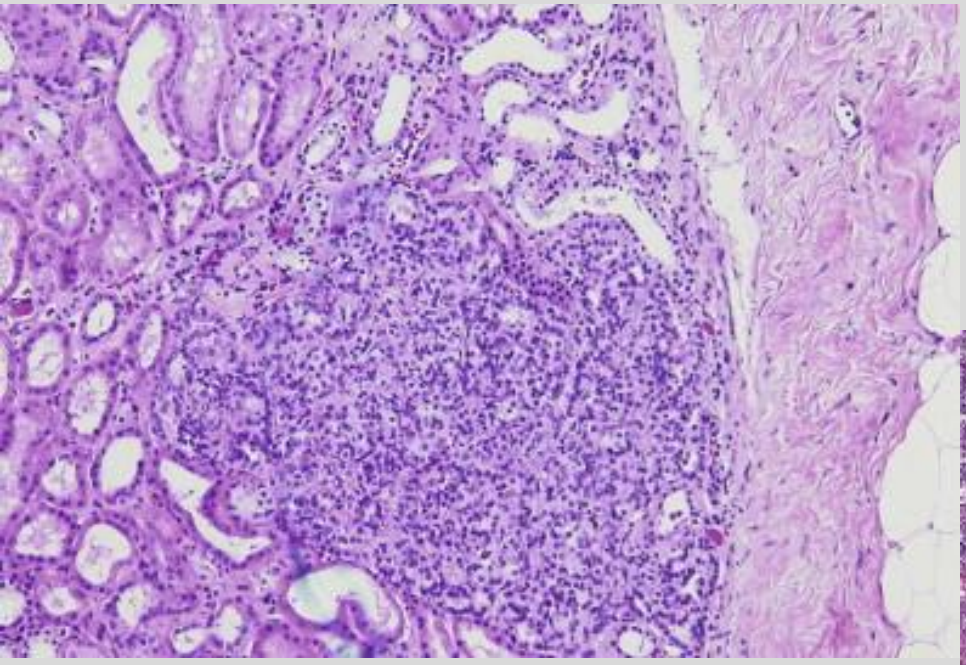
- Benign tumor of renal tubular epithelium
- Usually papillary: “papillary adenoma”
  - Another rare variety: “metanephric adenoma”
- Small, cortical, well-circumscribed
  - **Usually less than 1.5cm** (varies by source**); size cutoff is >3 cm**
- Clinical:
  - Uncertain malignant potential (treat as carcinoma)
- Gross:
  - Pale yellow-gray, discrete, well-circumscribed nodules
- Histology:
  - Complex, branching, papillomatous structures with numerous complex fronds
  - Cells are cuboidal to polygonal with small, central nuclei, little cytoplasm, no atypia

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Papillary Adenoma, Histology

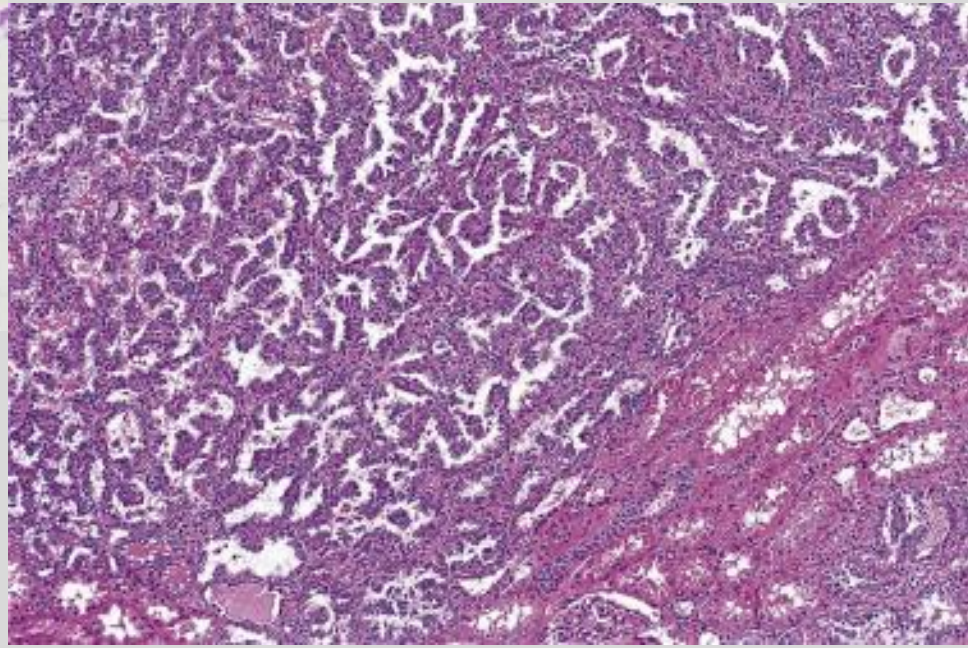
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Located in **renal cortex**; clearly distinguishable



Small, central nuclei, little cytoplasm, no atypia

Complex, branching papillomatous structures



Practical Renal Pathology, 2013; Diagnostic Histopathology of Tumors

Angiomyolipoma

- **Benign tumor** originating from perivascular epithelioid cells
- Composed of **blood vessels, smooth muscle, and adipose tissue**
- Sporadic (more common in women)
  - Angiomyolipoma cells express estrogen receptor
- Clinical features
  - Mostly asymptomatic
  - Spontaneous retroperitoneal hemorrhage
  - Hematuria
  - Impaired renal function
- Increased frequency in **tuberous sclerosis (TS)**
  - LOF in ***TSC1*** (Hamartin, Chr 9) and ***TSC2*** (Tuberin, Chr 16) tumor suppressor genes
  - Interact to inhibit activity of mammalian target of rapamycin (**mTOR**), a kinase that stimulates cell growth
  - Occur in 25-50% of TS cases

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Tuberous Sclerosis

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Angiofibromas of Tuberous Sclerosis

Tuberous Sclerosis

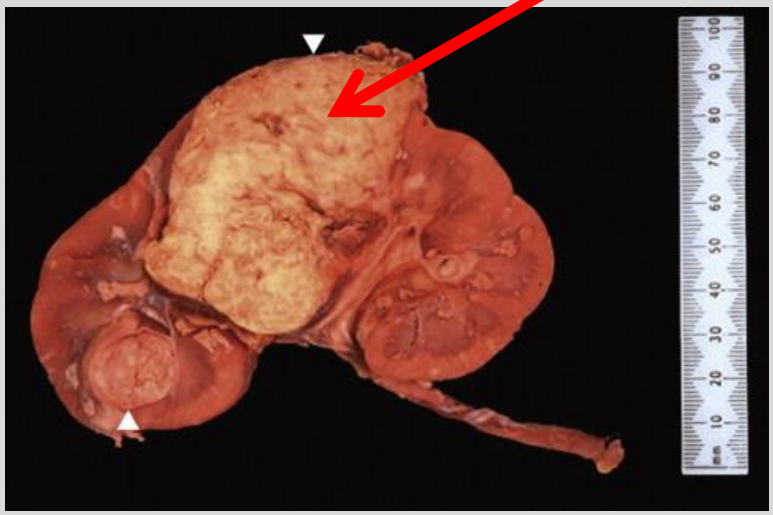
- Cardiac rhabdomyomas
- Renal angiomyolipomas
- Retinal glial hamartomas
- Cutaneous angiofibromas

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Angiomyolipoma, Gross pathology

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Yellowish-tan color



Multifocal

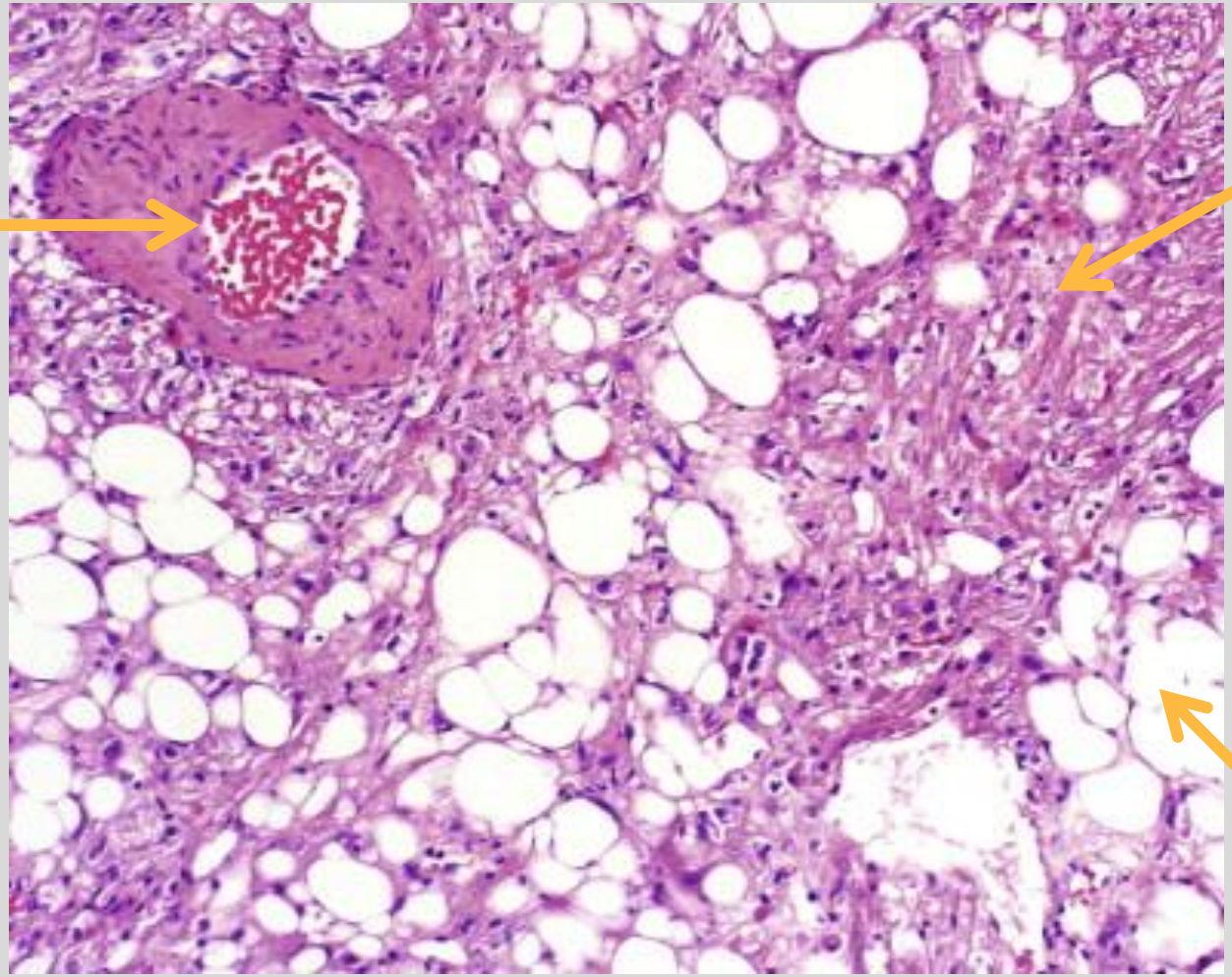


Almost always hyperechoic on renal u/s, sharply demarcated lesion in renal pelvis

radiopaedia.org; Robbins & Cotran Pathologic Basis of Disease, 10th edition- Chapter 20: The Kidney

Angiomyolipoma, Histopathology

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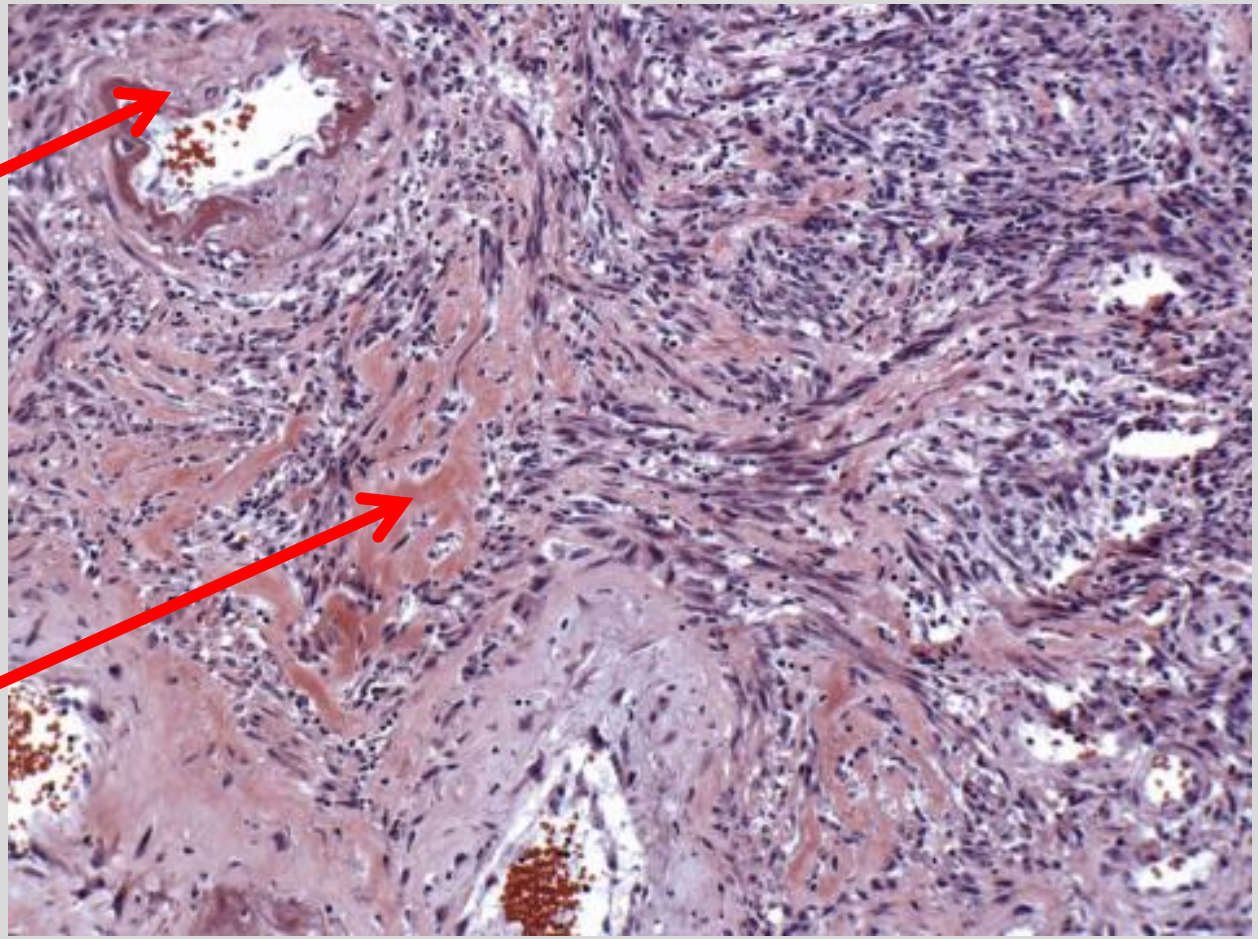


Practical Renal Pathology, 2013

Angiomyolipoma, Histopathology

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H&E, 200x



Thick-walled, artery-like blood vessels

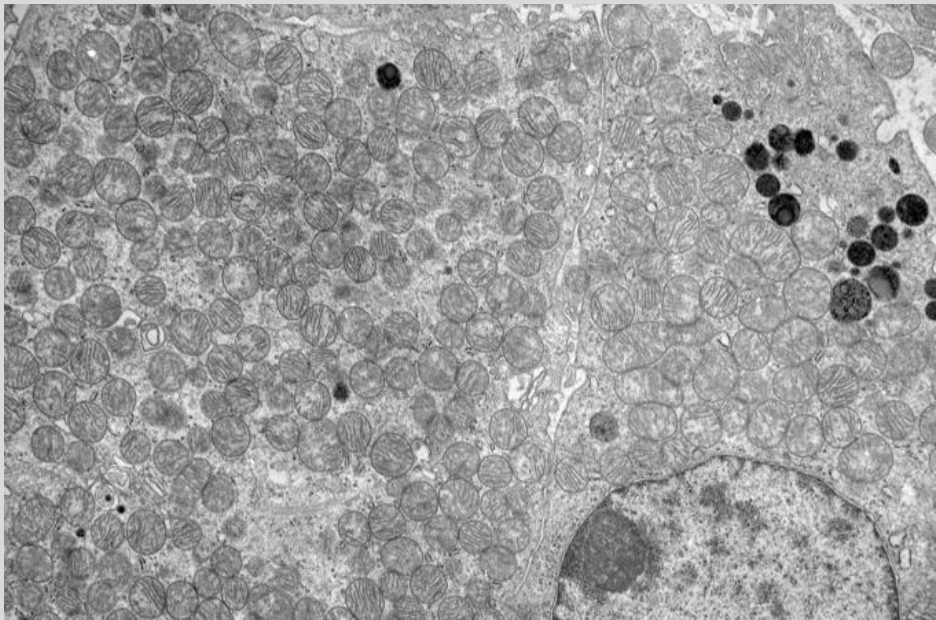
Smooth muscle cells

(Less adipose in this preparation)

Diagnostic Atlas of Renal Pathology, 3e

Oncocytoma

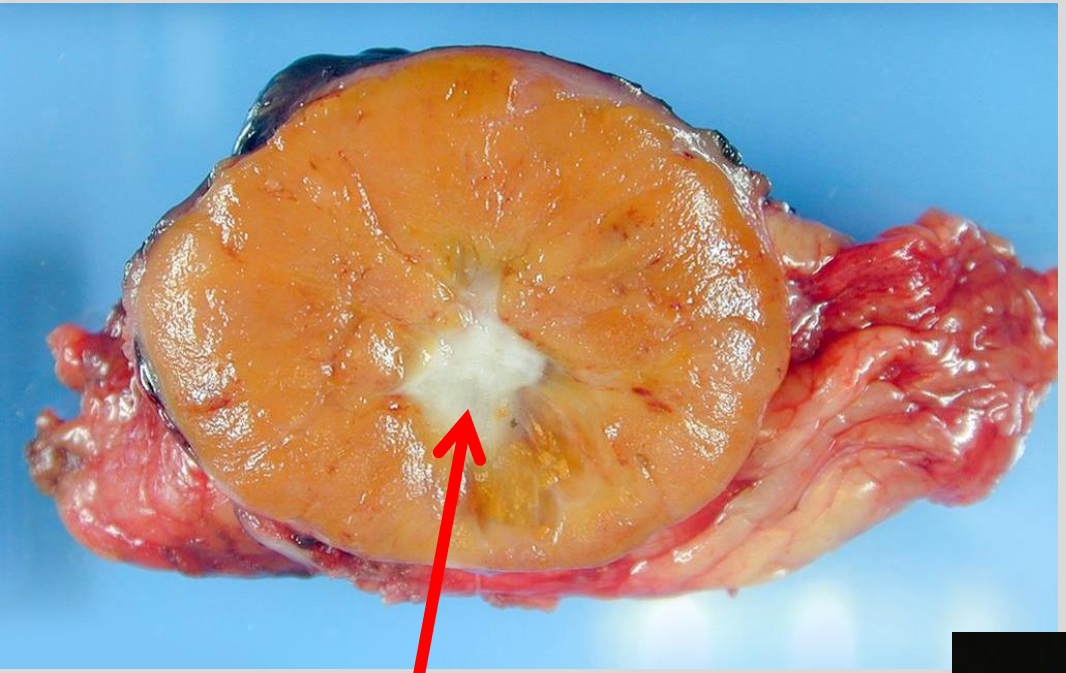
- Benign tumor
  - **Rarely metastasizes**
- 5-10% of resected renal neoplasms
  - Male > female (slightly)
- Thought to arise from **intercalated cells** of the collecting ducts
- Gross:
  - Well-circumscribed, homogenous, encapsulated, **mahogany-brown**
  - **Central scar** may be present
- Histology:
  - Granular, eosinophilic cytoplasm
  - Uniform round nuclei, central nucleoli
  - Many **mitochondria (EM)**



Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney

Oncocytoma, Gross Pathology

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Well-circumscribed
Mahogany-brown color

Central scar



https://radiopaedia.org/cases/renal-oncocytoma-gross-pathology-2; Diagnostic Atlas of Renal Pathology

Oncocytoma, Histopathology

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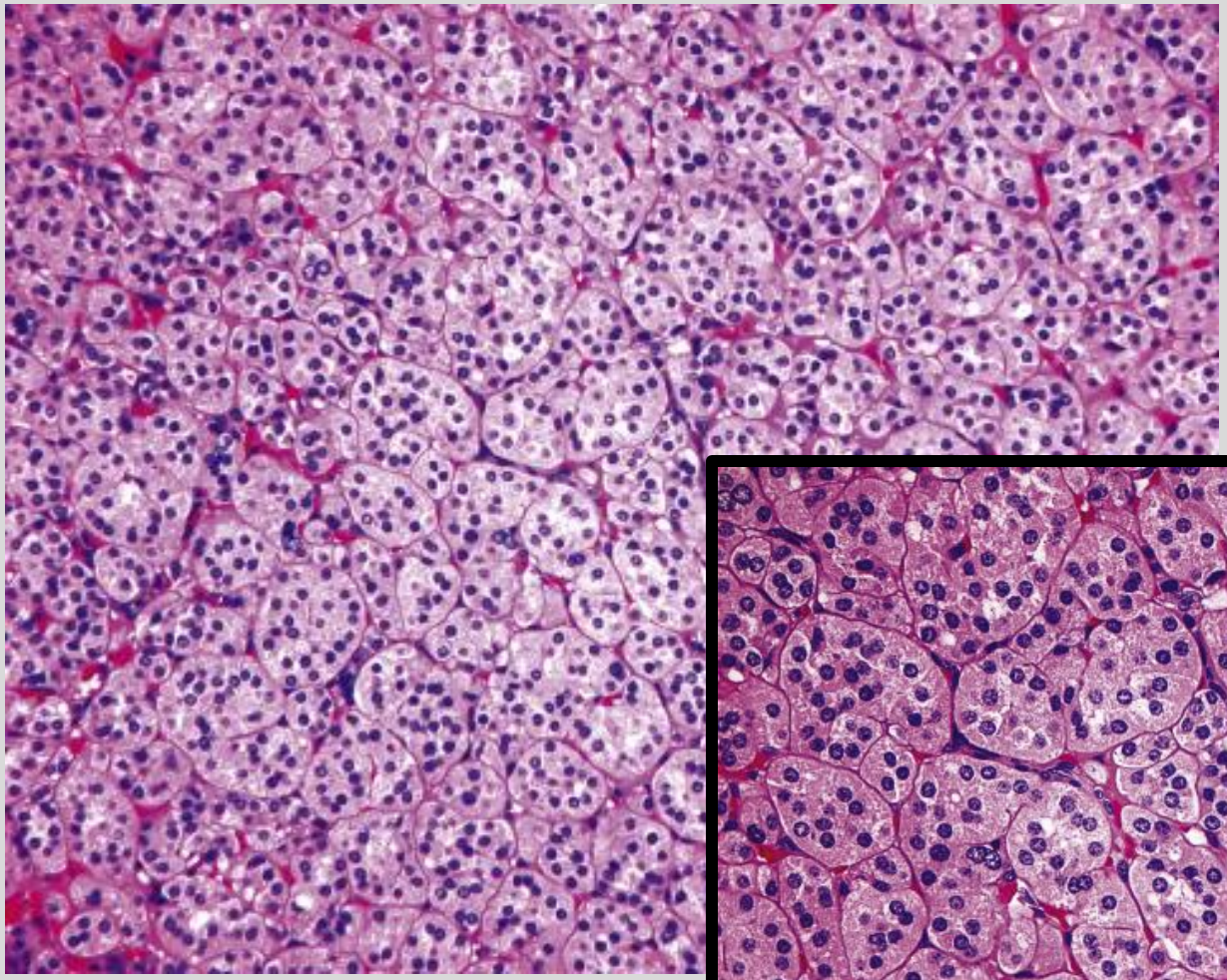
Granular, red cytoplasm

Uniform round nuclei

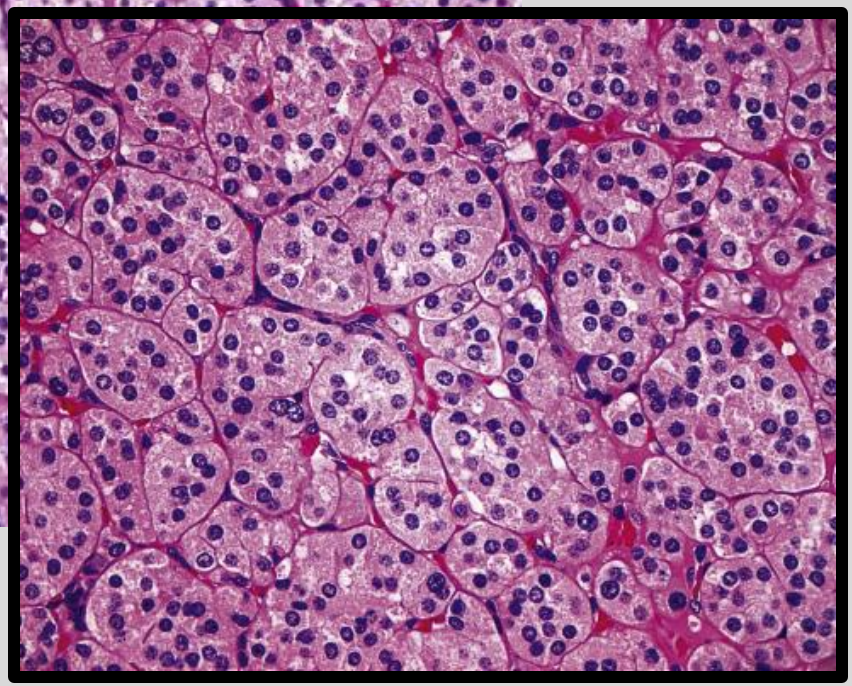
Central nucleoli

Homogenous pink cells

H&E, 100x



H&E, 400x



Diagnostic Atlas of Renal Pathology

Renal Cell Carcinoma (RCC)

- **Malignant tumor:** Represents 3% of all newly diagnosed cancers in the U.S. and account for 85% of renal cell cancers in adults; most common primary cancer in kidney
- Mostly **sporadic**; can be inherited
  - 5% are familial - More likely to be bilateral and arise younger (adults)
- Occurs in older individuals (average ~60yrs)
  - Male > female
  - **Tobacco use** is a significant risk factor
  - Additional risk factors: obesity, HTN, unopposed estrogen therapy, asbestos/petroleum/heavy metal exposure
  - Inc. risk in pt. with ESRD, CKD, acquired cystic disease and tuberous sclerosis
- Arises from **renal tubules**

Renal Cell Carcinoma - continued

- **Five subtypes:**
  - **Clear cell (most common)**
  - **Papillary**
  - **Chromophobe**
  - **Xp11 translocation**
  - **Bellini duct aka Collecting duct**
- Tends to invade **renal vein (may extend into IVC)**
- Tends to metastasize widely and early (before s/s)
- 15% of patients have evidence of metastases at time of presentation (most commonly to lungs and bone followed by regional lymph nodes, liver, adrenal glands, and brain)
- 5-year survival = 45%; without metastasis = 70%
- Treatment: partial or radical nephrectomy depending on tumor location/size (<4 cm)/comorbidities (e.g., DM); drugs that inhibit VEGF (e.g., Bevacizumab) and various tyrosine kinases for metastatic

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Renal Cell Carcinoma – Familial

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- 5% Familial/Syndromic variants
  - More likely to be bilateral and arise earlier (younger adults)
- **Von Hippel-Lindau (VHL) syndrome**
  - Renal cysts and bilateral clear cell RCC (**ccRCC**)
  - **Hemangioblastomas** of cerebellum and retina
  - **Pheochromocytoma** (tumor of the adrenal medulla)
  - VHL gene, **AD (3p)**
- **Tuberous Sclerosis**
  - Autosomal Dominant
  - *TSC1* – Hamartin (9q) and *TSC2* – Tuberin (16p)
  - Angiomyolipoma is more common in TS than ccRCC
- **Hereditary Papillary Carcinoma**
  - Autosomal Dominant
  - Mutations in **MET** proto-oncogene
  - Presents with multiple bilateral tumors

*__VHL__ and __TS__ are both associated with **hereditary clear cell carcinoma**

Hereditary RCC Overview

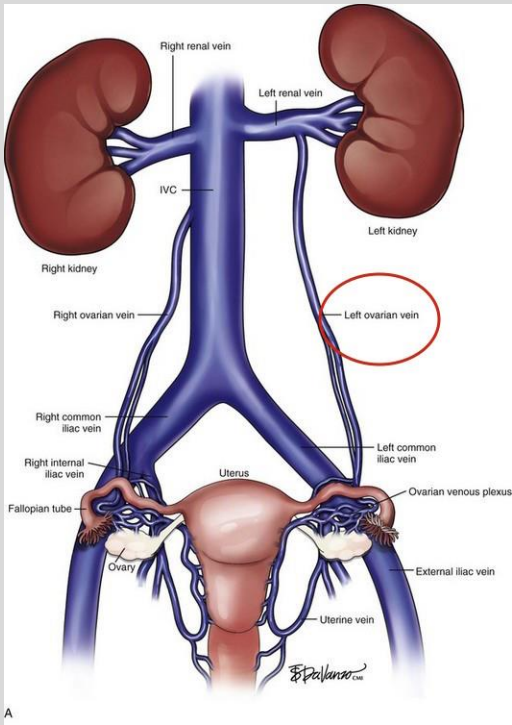
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Syndrome	Gene	Histology	Inheritance Pattern	Other systems Involved
Von Hippel-Lindau	VHL gene (chr. 3p) →overactivation of HIF → angiogenesis, cell growth, cell metabolism	Clear cell carcinoma	Autosomal dominant	Renal cysts and bilateral clear cell RCC (ccRCC) **Hemangioblastomas of cerebellum and retina** **Pheochromocytoma** (tumor of the adrenal medulla)
Tuberous sclerosis	TSC1 (chr. 9) & TSC2 (chr. 16): LOF in TSC1/TSC2 tumor suppressor genes → inhibit mTOR (kinase that stimulates cell growth)	Clear cell carcinoma (also angiomyolipoma which is more common in TS)	Autosomal dominant	Cardiac rhabdomyomas Renal angiomyolipomas Retinal glial hamartomas Cutaneous angiofibromas
Hereditary Leiomyomatosis and renal cell cancer syndromes	FH gene	Papillary	Autosomal dominant	Leiomyomas of skin and uterus PET-positive adrenal adenomas
Birt-Hogg-Dube Syndrome	BHD gene	Chromophobe/hybrid oncocytoma	Autosomal dominant	Cutaneous fibrofilliculoma or trichodiscoma Pulmonary cysts Spontaneous pneumothorax
Hereditary Papillary carcinoma	MET protooncogene AKA hepatocyte growth factor-mediates growth, cell mobility, invasion, and morphogenetic differentiation	Papillary	Autosomal dominant	N/A

Adapted from JNCNN.org & Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney

Renal Cell Carcinoma, Clinical

- **Symptoms:** rarely see all of these
  - **Costovertebral/flank pain** or abdominal pain
  - Palpable flank or abdominal mass
  - **Hematuria** (intermittent/microscopic)
  - Sometimes constitutional symptoms (e.g. fever)
- **Paraneoplastic syndromes ("PEAR"-aneoplastic)**
  - PTHrP: Hypercalcemia
  - Ectopic EPO: Polycythemia
  - ACTH: Cushing syndrome from steroid hormone or ACTH release
  - Renin release: hypertension
- Left-sided varicocele in males (rare)
  - Tumor blocks left renal vein (drainage pathway of left testicular vein)



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Clear Cell RCC (ccRCC)

- **Most common** type of RCC (70-80%)
- Unilateral, more common in upper pole of kidney
- Originates in the **proximal tubule**
- Loss of short arm of **chromosome 3**
  - Sporadic (95%), familial or associated with VHL syndrome
  - VHL tumor suppressor gene is within chromosome 3
  - VHL targets transcription factor hypoxia inducible factor 1 (HIF-1) → elevated HIF-1 → promotion of angiogenesis (VEGF), cell growth (IGF-1), reprograms cell metabolism (MYC)

Histology:

- Round-polygonal cells with clear cytoplasm (due to abundance of lipids/glycogen)
- Non-papillary
- Delicate branching vasculature (**"chicken-wire" pattern**)

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In this case, the tumor extended up the vena cava and occluded the adrenal vein, leading to hemorrhagic adrenal infarction in the adrenal at the top of the specimen



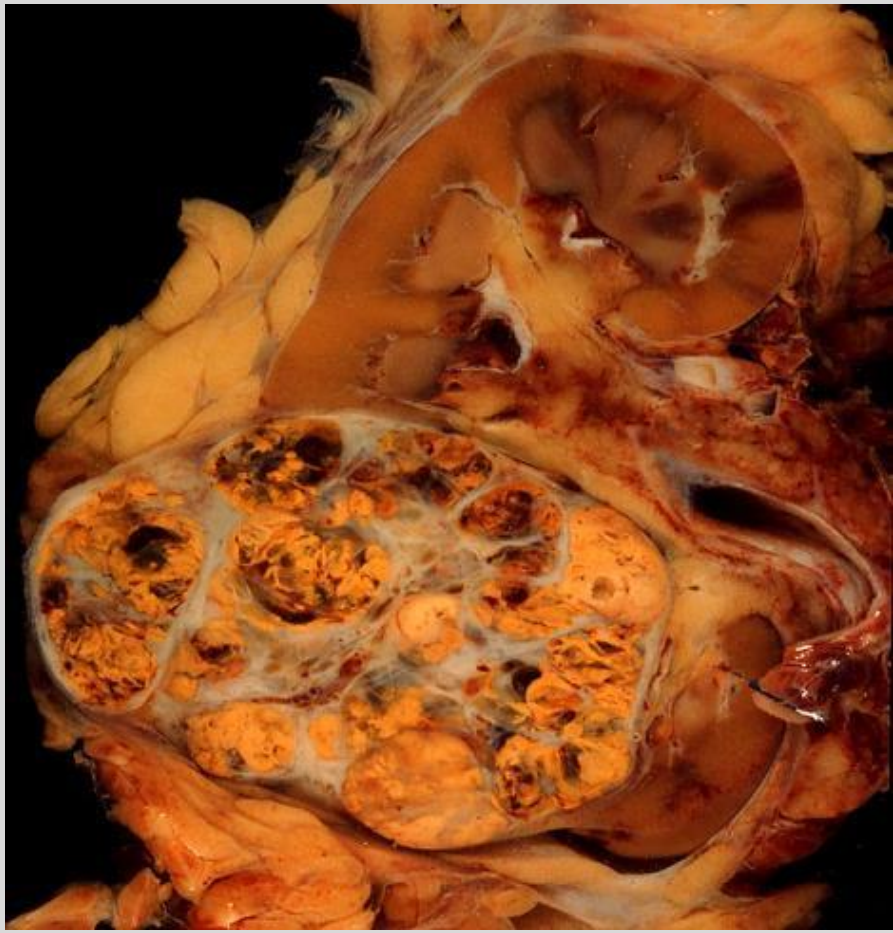
Renal cell carcinomas have a tendency to invade into the renal vein, (white arrow)

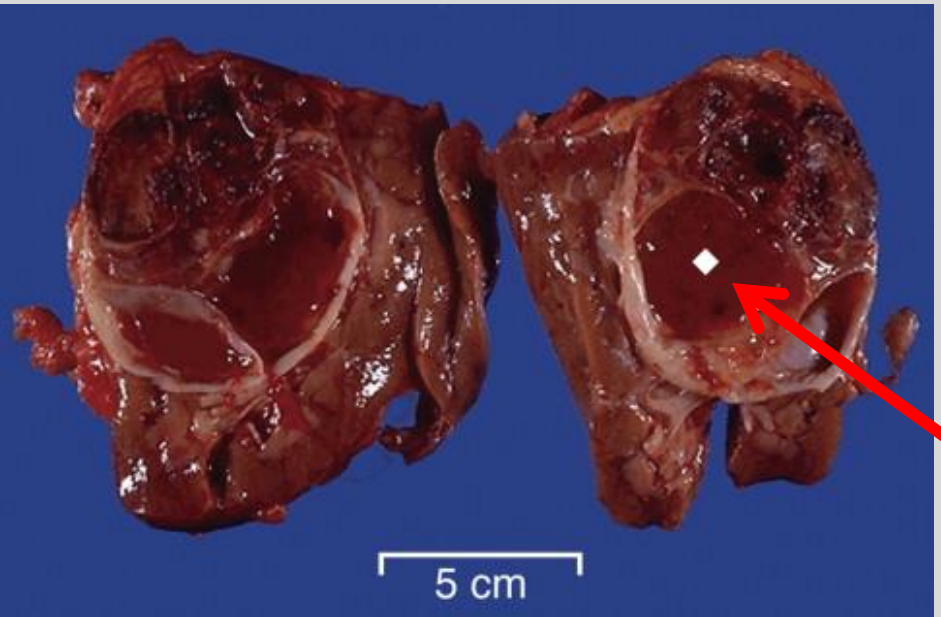
Webpath.med.Utah.edu

ccRCC, Gross Pathology

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- Bright yellow (due to lipid in tumor cells) to grey-white mass which distorts the kidney
- Areas of hemorrhage and necrosis are common, as well as **cystic change**
  - Differential diagnosis of other cystic conditions (e.g. ADPKD)

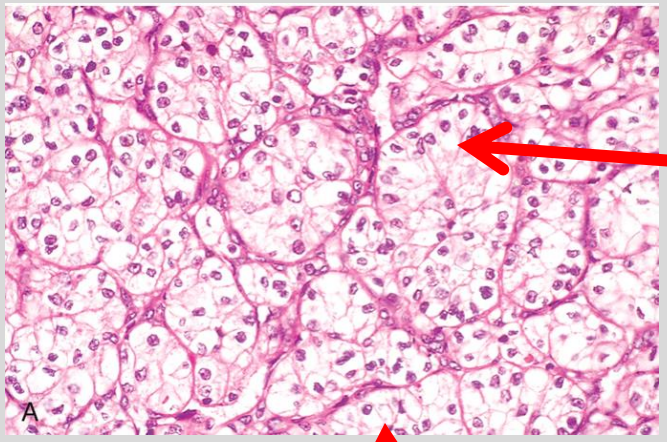




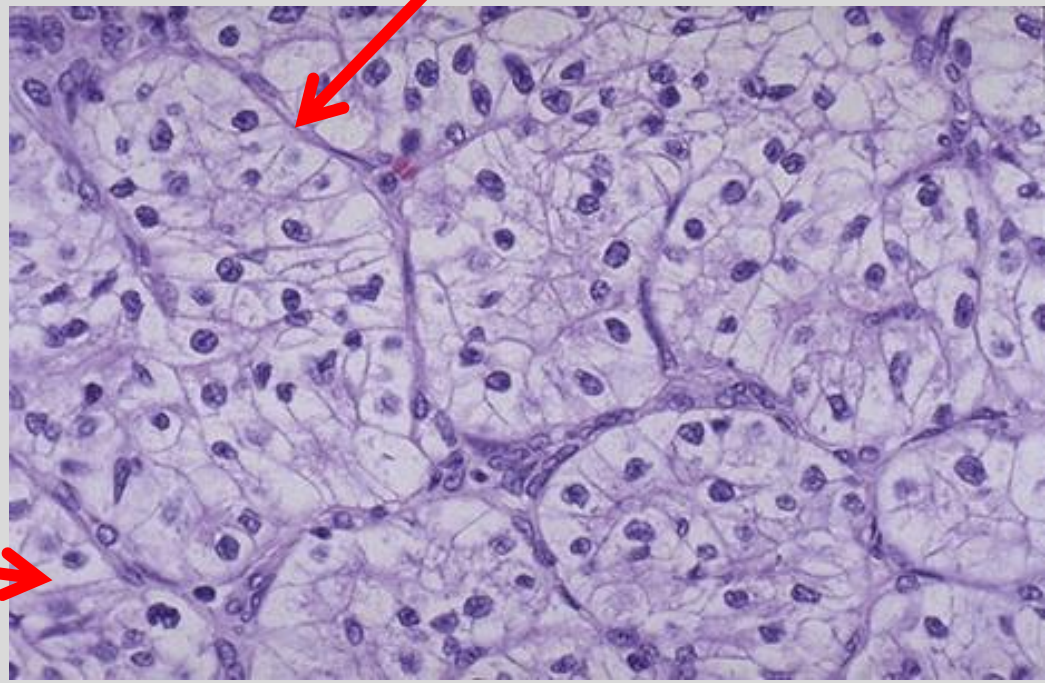
Cystic with extensive hemorrhage

ccRCC, Histopathology

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Delicate fibrovascular stroma
("chicken wire")



Clear cytoplasm

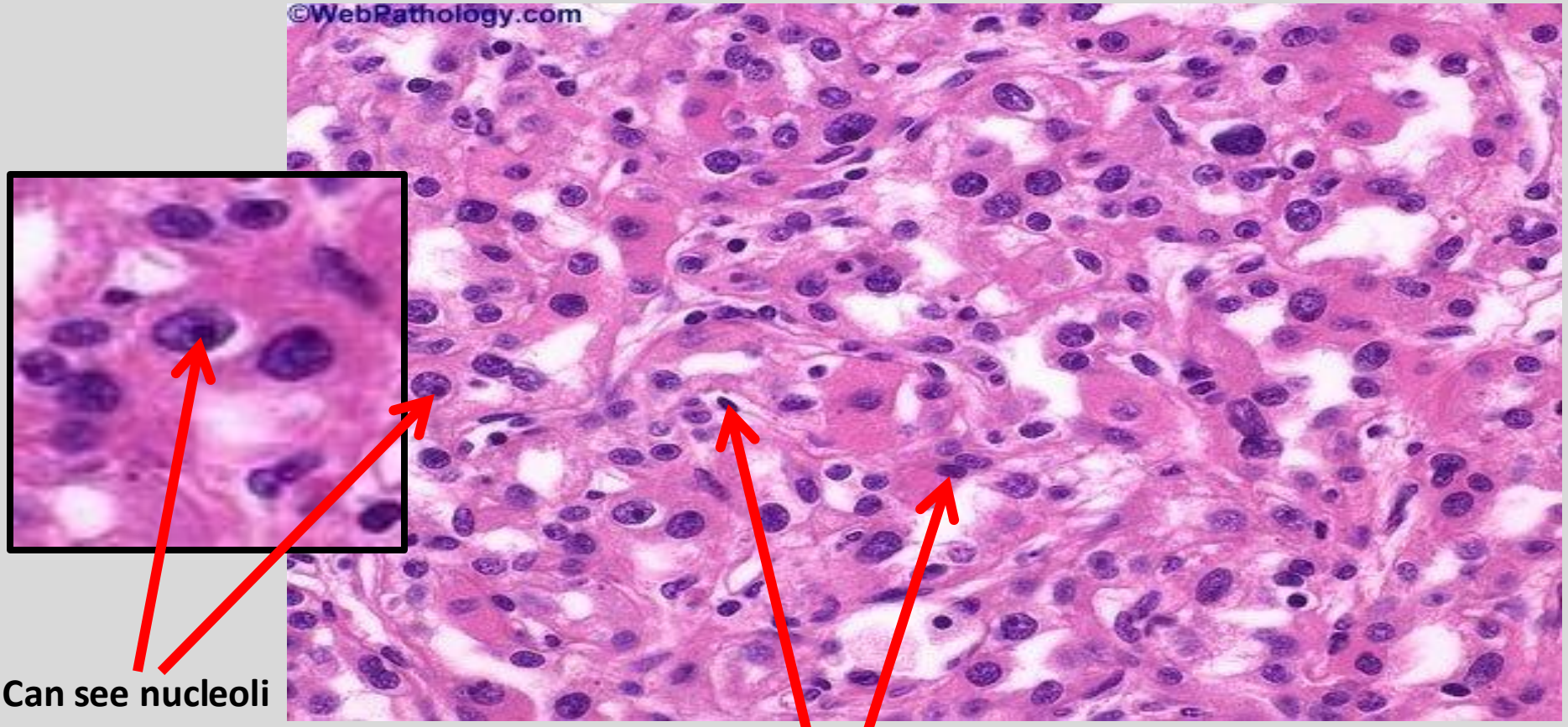
Fuhrman Grading

- **Higher grades correlate with more aggression**
- 1 – small, round, uniform nuclei; inconspicuous nucleoli, look like lymphocytes (rare)
- 2 – slightly irregular nuclei; see nucleoli at 40x only; open chromatin (40% of tumors)
- 3 – nuclei very irregular; see **nucleoli at 10x** (low power); open chromatin (30-40% of tumors)
- 4 – mitoses, bizarre, multilobulated, pleomorphic cells plus Grade 3 features; macronucleoli – may also be sarcomatoid (15% of tumors)
- ***Important predictor of survival in patients with localized Clear Cell Renal Cell Carcinoma**
  - **The higher the grade, the worse the prognosis (lower 5-year survival rate)**

Fuhrman Grade 3 (Example)

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(ccRCC)



Can see nucleoli at 10x mag.

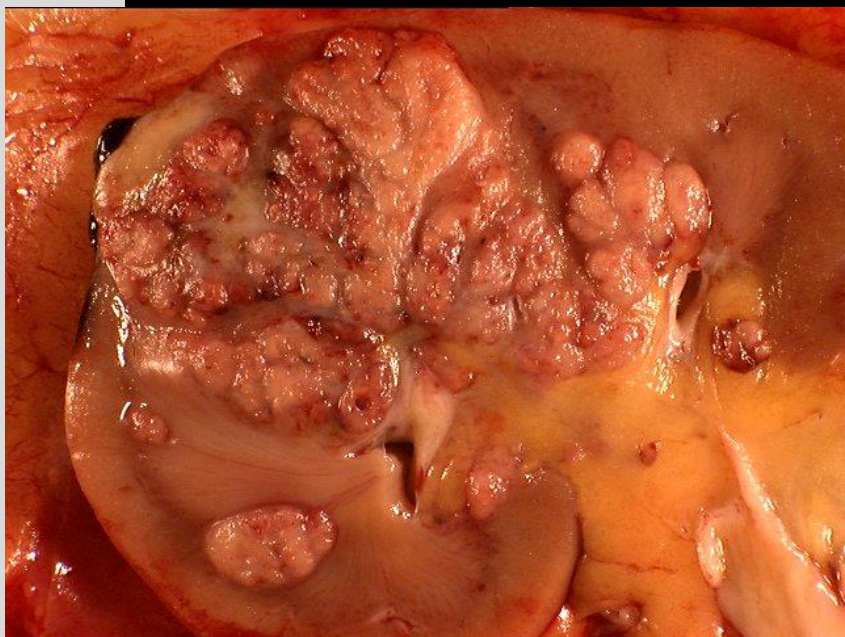
Irregular nuclei (not homogenous)

Papillary RCC

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- 10-15% of RCC cases (less than ccRCC)
- Usually **multifocal**
- Most common tumor in patients who develop **dialysis-associated cystic disease**
- **Multifocal, hemorrhagic,** and **cystic** on gross pathology
- Sporadic:
  - Associated with **Trisomies 7, 16, 17, and loss of Y chr** (males)
- Familial:
  - Associated with **Trisomy 7** (Chr 7 locus for **MET** proto-oncogene)
  - MET: AKA hepatocyte growth factor-mediates growth, cell mobility, invasion, and morphogenetic differentiation



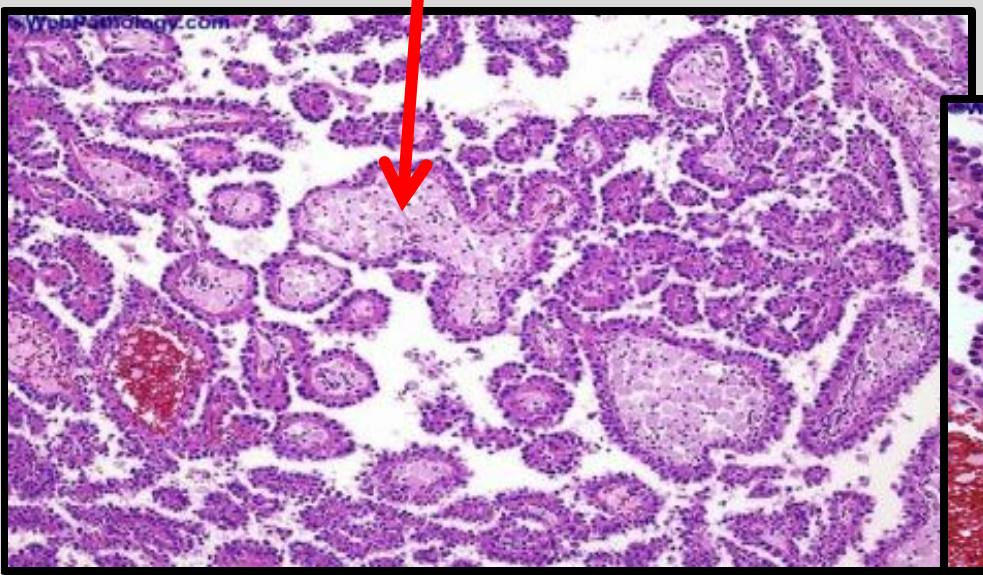


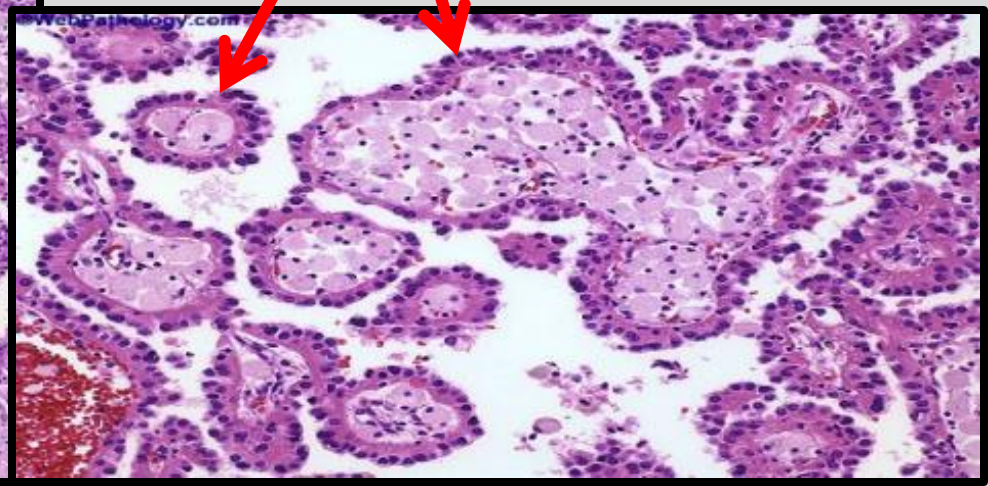
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Papillary RCC, Histopathology

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- Thought to arise from **distal convoluted tubules**
- Cuboidal or low columnar cells in **papillary formations**
- **Psammoma bodies** may be present
- **Foamy macrophages** within papillary cores





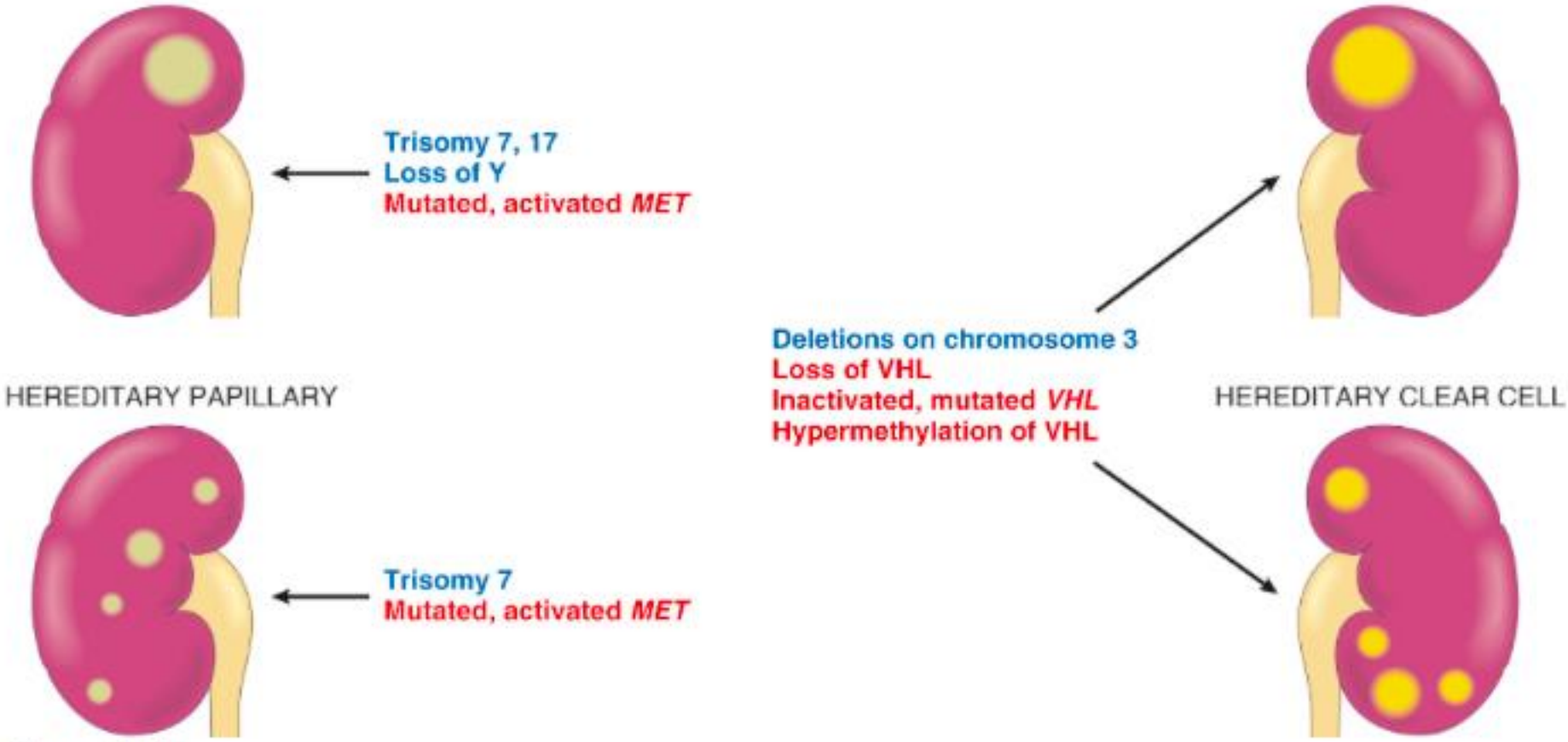

SPORADIC PAPILLARY
Trisomy 7, 17
Loss of Y
Mutated, activated MET
SPORADIC CLEAR CELL
Deletions on chromosome 3
Loss of VHL
Inactivated, mutated VHL
Hypermethylation of VHL
HEREDITARY PAPILLARY
Trisomy 7
Mutated, activated MET
HEREDITARY CLEAR CELL

Figure 20.49

Chromophobe RCC

- 5% of RCC cases
- Multiple chromosome losses
- Very **good prognosis** (least aggressive)
- Thought to arise from **intercalated cells** (like oncocytomas)
  - This is malignant, oncocytomas are benign
- Histology:
  - Prominent cell membranes
  - **Pale eosinophilic cytoplasm**
    - Often have halo around nucleus
    - **"vegetable cell wall"** appearance
  - Often hard to distinguish microscopically from oncocytomas

Well-circumscribed, globular, solid, tan or brown

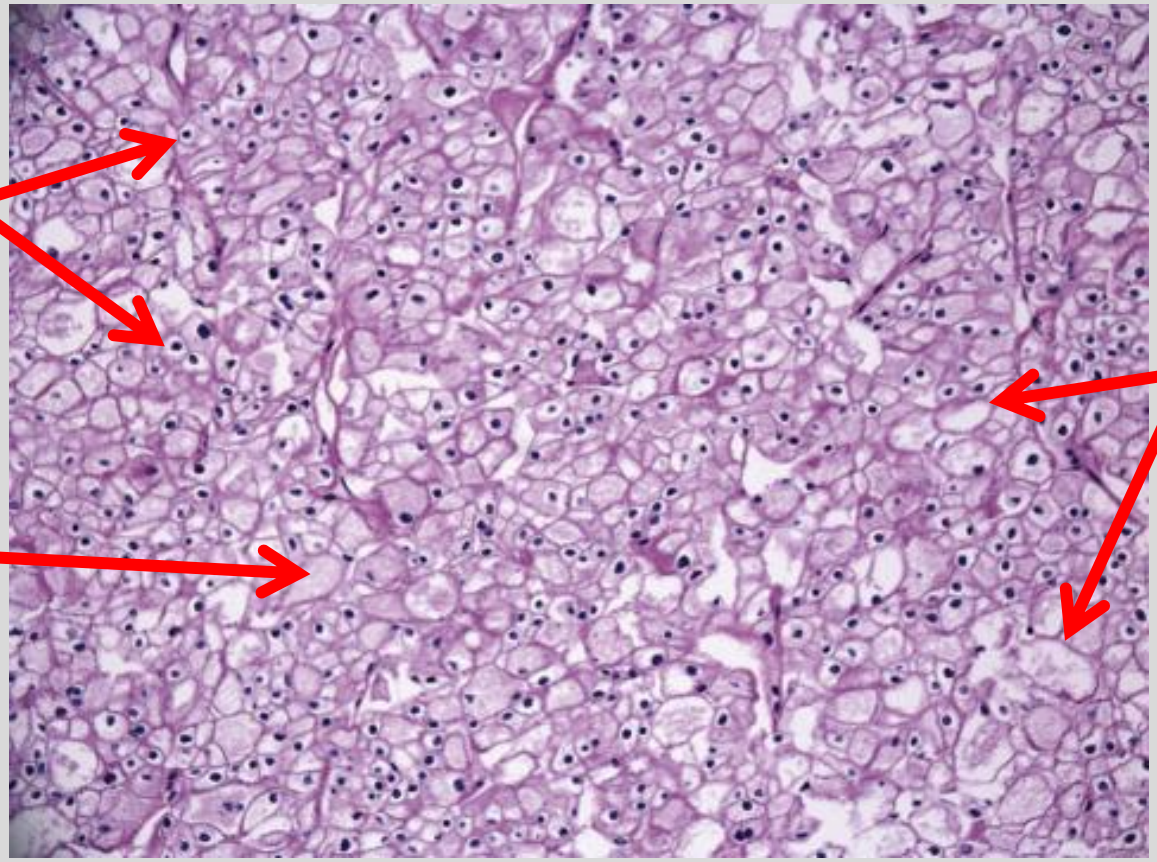


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Chromophobe RCC – Histopathology

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H&E, 100x



"Vegetable cell wall" appearance

Diagnostic Atlas of Renal Pathology, 3e

XP11 Translocation RCC

- Genetically distinct subtype of renal cell carcinoma
- Occurs in young patients
- Defined by translocations of TFE3 gene at XP11.2 → overexpression of TFE3 transcription factor
- Histology:
  - Clear cytoplasm with a papillary architecture

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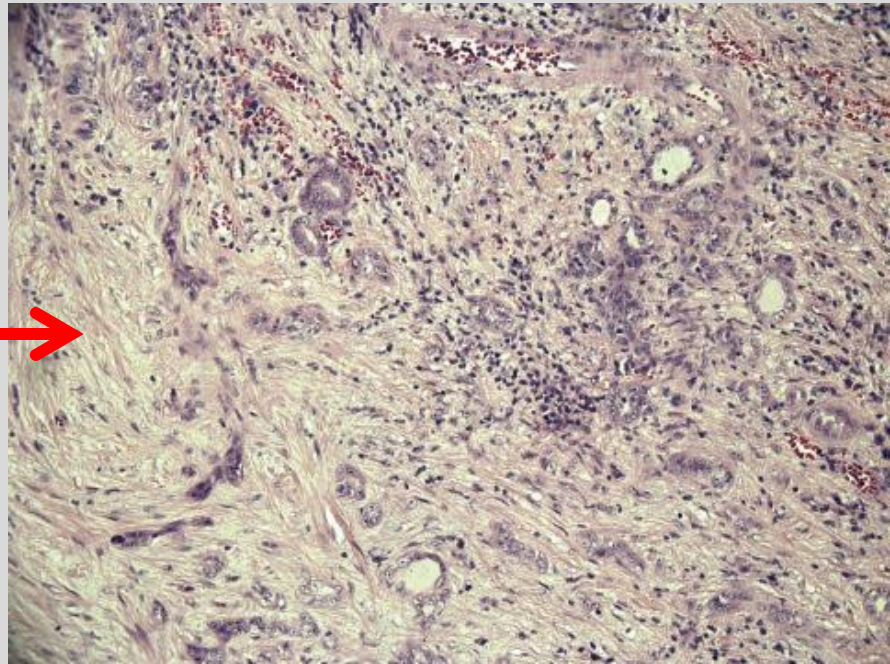
Collecting (Bellini) Duct RCC

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- <1% of RCC cases (least common)
- Arise from **collecting ducts** in medulla of kidney
- Clinical:
  - Age varies widely, but mostly <50 years old
  - **Poor prognosis** (50% mortality at 2 years)
    - Very aggressive tumor
- <u>Histology:</u>
  - No definite cytogenetic pattern yet
  - Nests of malignant cells within fibrotic stroma
  - Complex, ductlike tubular cells

Diagnostic Atlas of Renal Pathology, 3e

H&E, 400x

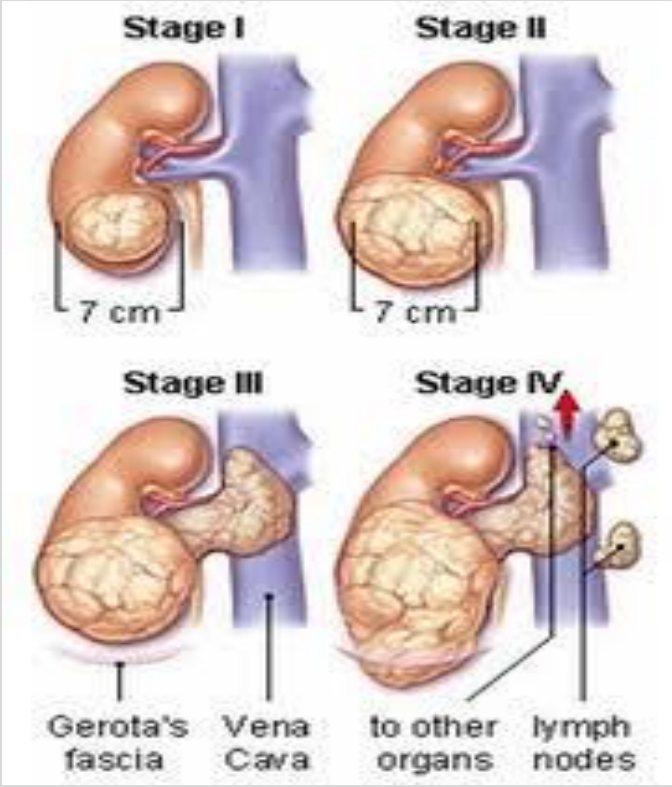


RCC – TNM Staging

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- **T** = Primary tumor
- **N** = Regional lymph nodes
- **M** = Distant metastases

- T1 – Tumor **≤ 7cm**, limited to kidney
- T2 – Tumor **> 7cm**, limited to kidney
- T3 – Tumor **extends into major veins and perinephric tissue**, but not into adrenal glad or beyond Gerota's fascia
- T4 – Tumor **extends beyond Gerota's fascia**
- N0 or N1
- M0 or M1



- Grades get added up into groups for staging
  - Different prognoses
  - Determines which type of treatment patient should receive

Wilms Tumor (Nephroblastoma)

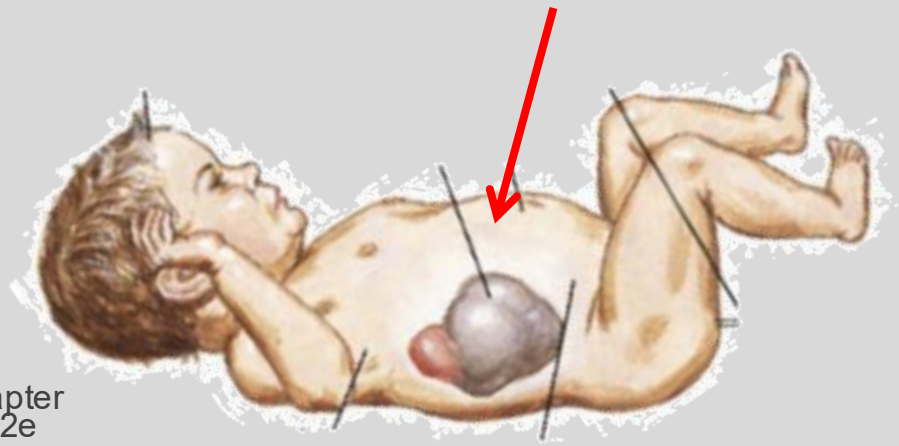
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- ***Most common primary renal tumor of childhood***
  - 4th most common **pediatric** malignancy in the U.S.
  - Peak age: between 2-5 years old
- 3 chromosomal disorders associated with increased risk:
  - **WAGR syndrome:** deletion of **11p13** (***WT1*** gene)
    - **W**ilms tumor (33% chance), **A**niridia, **G**enital anomalies, intellectual disability (formerly mental **r**etardation, inappropriate term: **R**ange of developmental delays)
    - Autosomal dominant
  - **Denys-Drash syndrome:** ***WT1*** gene → 90% risk of Wilms
    - Gonadal **dysgenesis** and early-onset nephropathy/renal failure (**diffuse** mesangial sclerosis)
  - **Beckwith-Wiedemann syndrome (BWS):** deletion of **11p15**
    - Organomegaly, macroglossia, hemihypertrophy, omphalocele, and enlarged adrenals
- 90% of cases are **sporadic** (nonsyndromic) and unilateral tumors
  - 5-10% are bilateral and have a median onset 10 months earlier than unilateral. These patients are assumed to harbor germline mutations.

Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney

Wilms Tumor – Clinical Features

- Large abdominal mass
- Hematuria, hypertension (due to renin release), intestinal obstruction
- DDX: Neuroblastoma
- Pulmonary metastasis can be present at time of diagnosis
  - Other common sites are local nodes and liver
- **Good prognosis** with **chemotherapy and resection**
- Possible association with:
  - Sarcoma
  - Leukemia, lymphoma
  - Brain tumors
  - GU tumors



Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney; Image: Netter's Illustrated Human Pathology, 2e

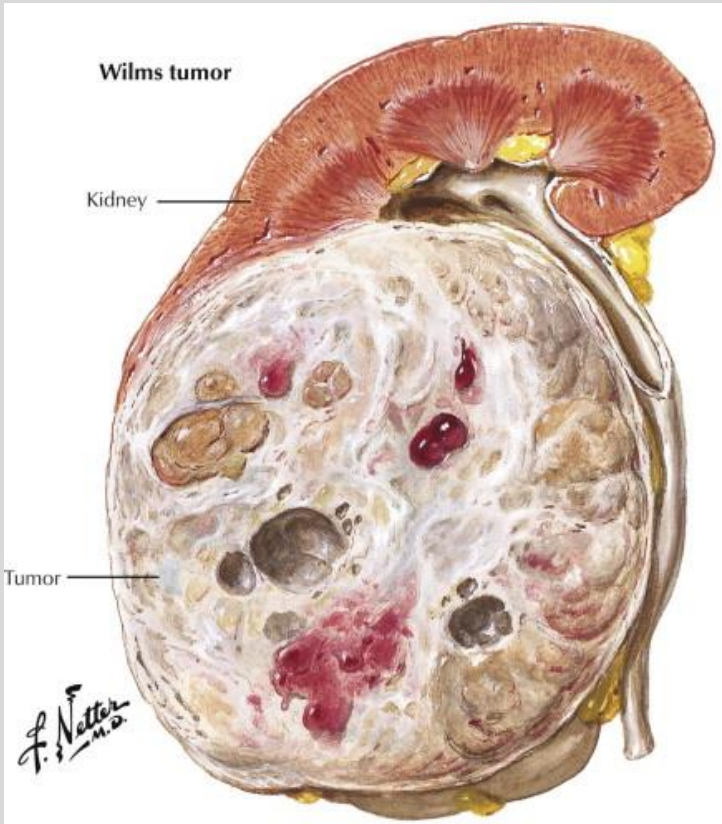
Wilms Tumor – Gross Pathology

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- Large, single **soft homogenous mass**
- **Tan-to-grey** color
  - May have foci of hemorrhage, necrosis, cyst formation
- Well-circumscribed margins with thin fibrous pseudocapsule



Robbins & Cotran Pathologic Basis of Disease, 10th edition-Chapter 20: The Kidney; Netter's Illustrated Human Pathology, 2e



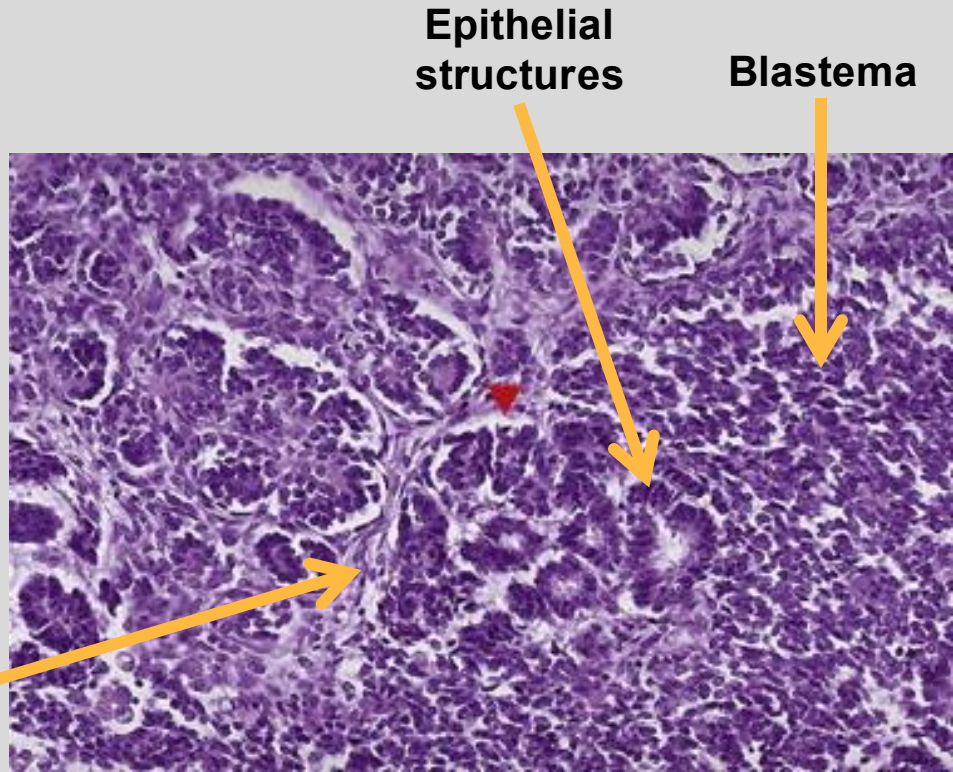
Wilms tumor. With pseudocapsule and characteristic variegated structure

Wilms Tumor – Histopathology

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- **Recapitulates** different stages of **nephrogenesis**
- **Triphasic histology** of:
  - Blastema (immature mesenchyme)
  - Stromal component
  - Epithelial structures
- Stromal component:
  - Usually fibrocytic or myxoid
  - Can have skeletal muscle, adipose tissue, cartilage, bone differentiation

**pediatric neoplasms are often composed of cells that resemble primitive embryonic counterparts: -blasts

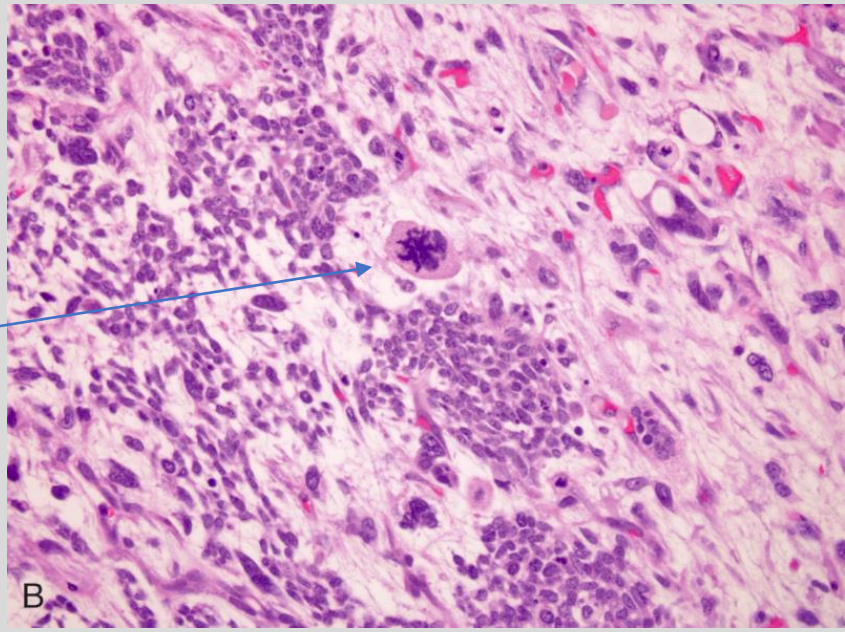


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Wilms Tumor, Histopathology

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- Focal anaplasia in 5% of Wilms tumors characterized by cells with hyperchromatic, pleomorphic nuclei, and abnormal mitoses.
- Anaplasia correlates with TP53 mutations and emergence of resistance to chemotherapy.



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Wilms Tumor – Histopathology

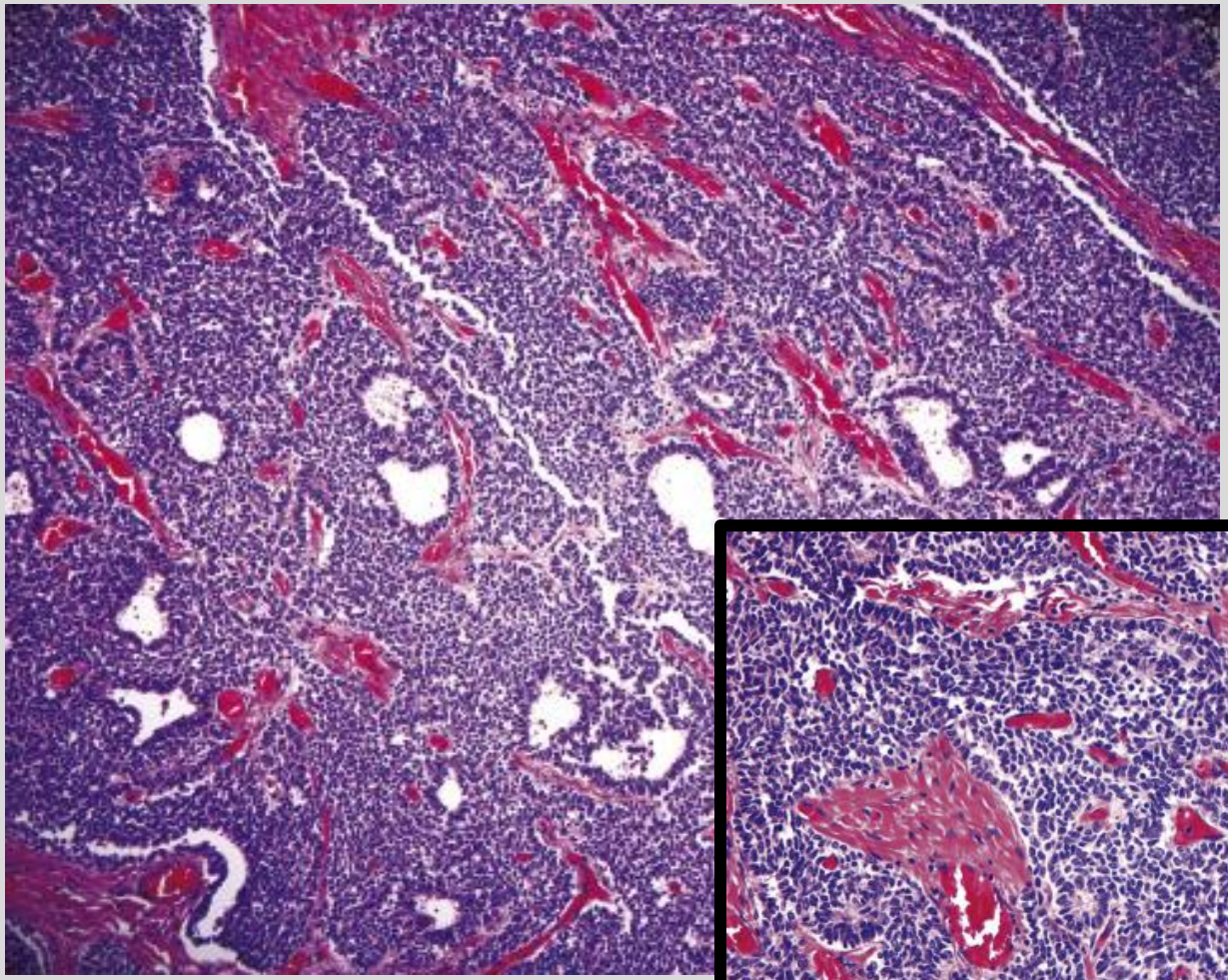
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Blastema: sheets of uniform small blue cells (densely packed)

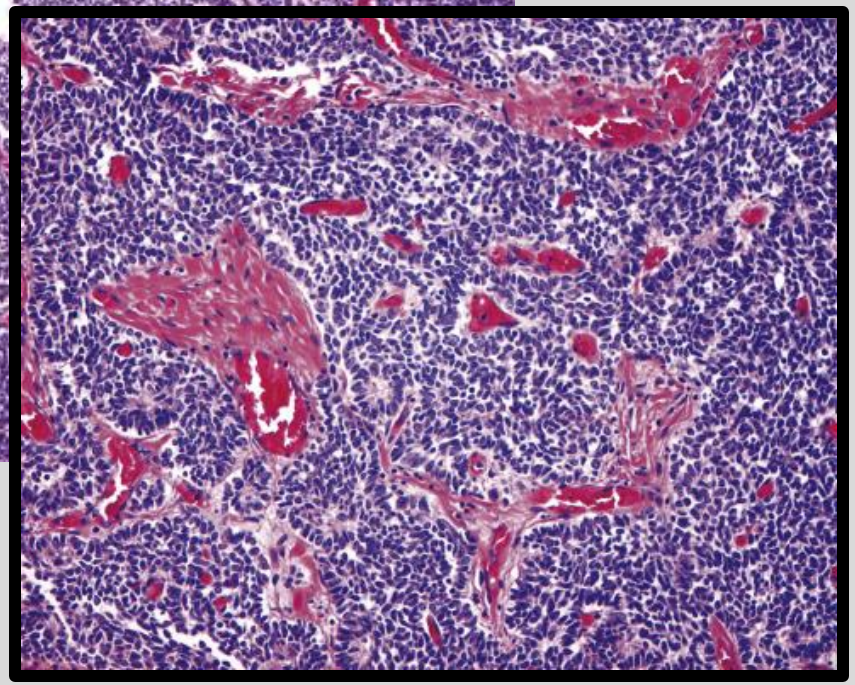
Epithelial: poorly differentiated tubular structures

Stroma: fibrovascular (contains fibroblasts and smooth muscle cells)

H&E, 100x



H&E, 200x



Robbins & Cotran Pathologic Basis of Disease, 10th edition- Chapter 20: The Kidney

Transitional Cell Carcinoma (TCC)

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- aka **Urothelial Carcinoma**
- 90% of bladder tumors, ~5% of renal tumors
  - Renal → appears in **renal pelvis and calyces,** up to distal urethra (covered by transitional epithelium)
- Chromosome **9p deletion** in superficial papillary and noninvasive *in situ* lesions; Chromosome **17p deletions** and **p53 mutations** in invasive TCC
- Male > female, 50-80 years of age
- Risk factors: cigarette smoking, exposure to polycyclic aromatic hydrocarbons or aromatic amines, long-term analgesic use, radiation, family history (first-degree relative)
- Clinical:
  - **Painless** hematuria present in over ¾ cases

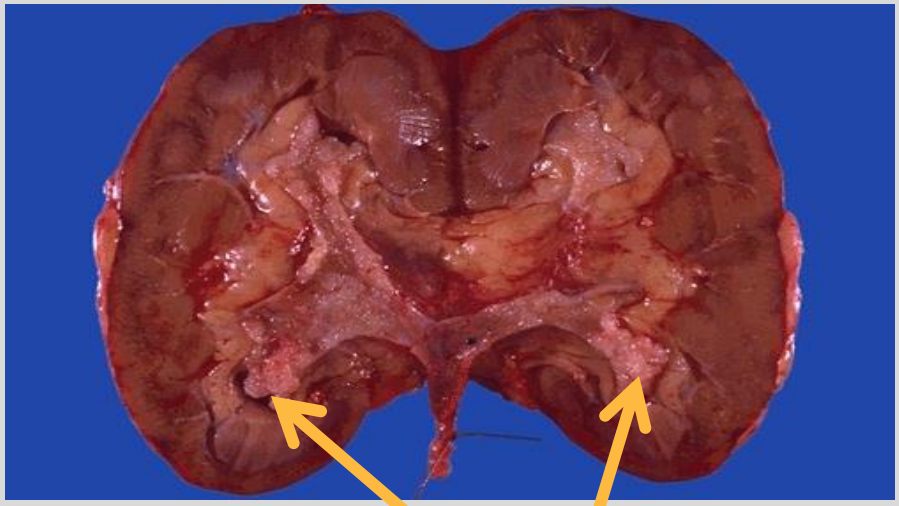


**You will learn more about this disease in the 'Pathology: Lower Urinary Tract' lecture*

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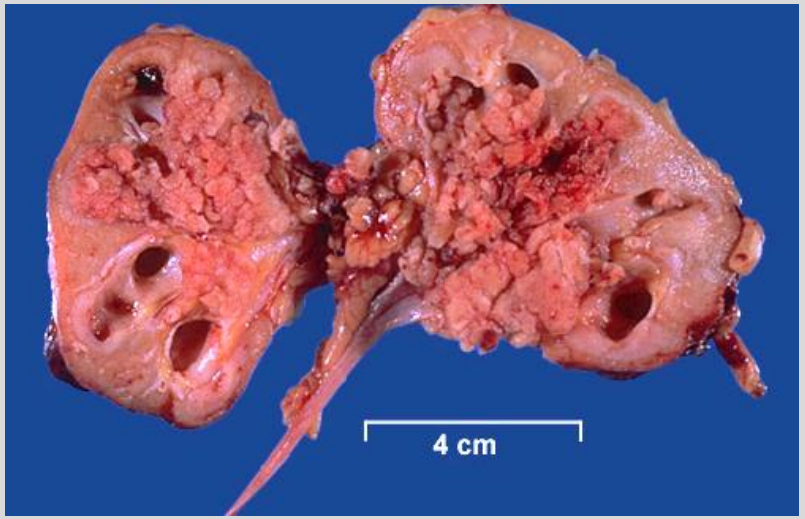
TCC (Renal) – Gross Pathology

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Tumors growing in renal pelvis and calyces

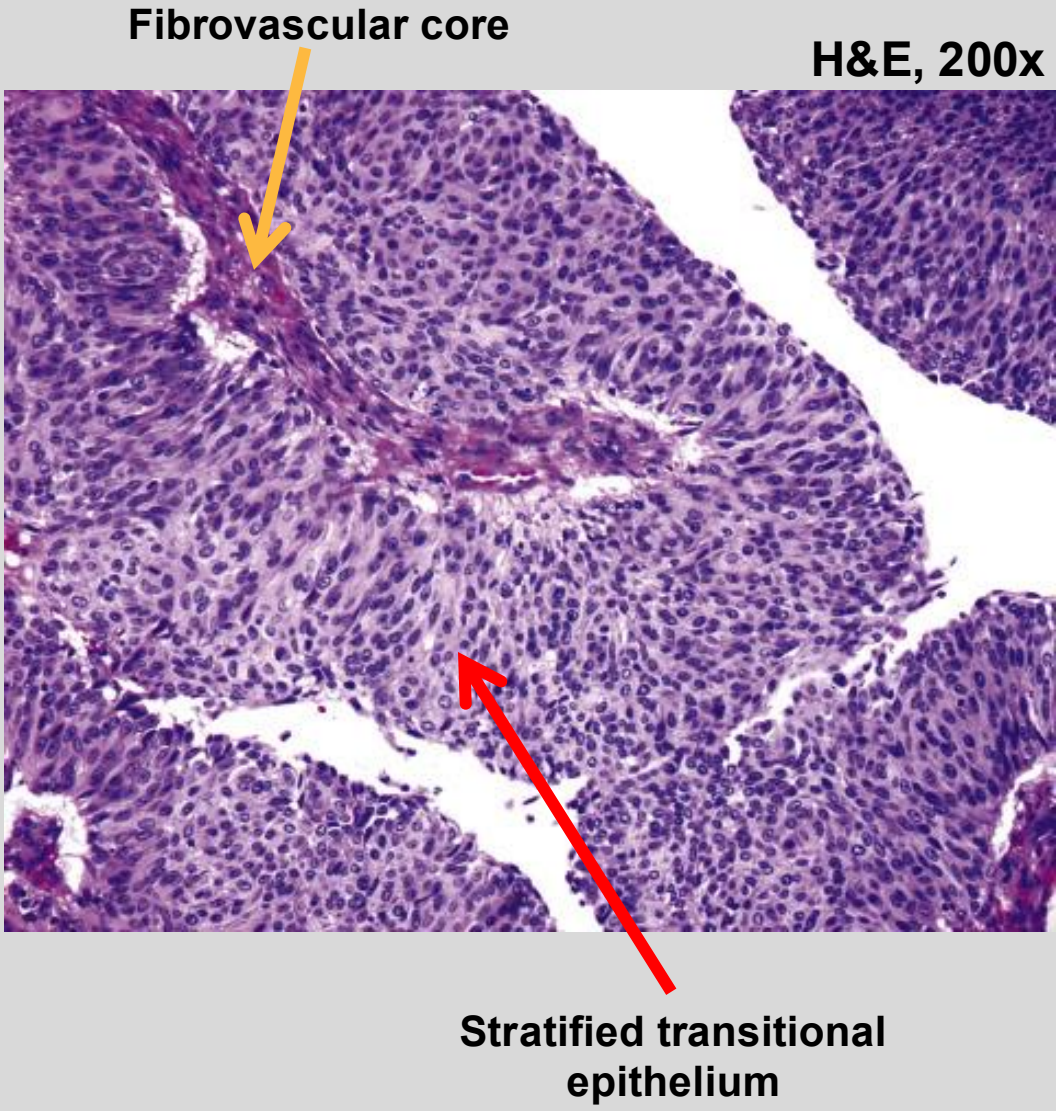
Aggressive tumor invading into renal parenchyma, causing obstruction and dilation of pelvis and calyces to produce hydronephrosis



TCC (Renal) – Histopathology

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- Epithelium has pleomorphic nuclei
  - Variable size and shape, as well as degree of differentiation
- Fibrovascular cores surrounded by stratified transitional epithelium
- Tumor appears similar to TCC of the bladder



Summary Chart

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Tumor	Cell & Location	Clinical & Associations	Gross	Histology
Papillary Adenoma	Renal cortex	Uncertain malignant potential	Small, well-circumscribed	Small central nuclei, little cytoplasm, papillomatous
Angiomyolipoma	Perivascular epithelioid cells	**TS**	Yellowish-tan	Thick-walled vessels, smooth muscle, adipocytes
Oncocytoma	**Intercalated cells**	Benign	Mahogany-brown, Central scar	Homogenous pink cells, many mitochondria
Clear Cell RCC	Proximal tubule	Smoking, VHL, TS; flank pain, hematuria	Colorful, hemorrhage, necrosis, cysts	**"chicken wire" pattern, clear cytoplasm**
Papillary RCC	Distal convoluted tubule	**Trisomy 7; Dialysis-associated cystic disease**	Multifocal, hemorrhagic, and/or cystic	**Foamy macrophages within papillary core**
Chromophobe RCC	**Intercalated cells**	Malignant, good prognosis	Well-circumscribed, solid, tan-brown	**"vegetable cell wall", pale cytoplasm, halo around nucleus**
Collecting Duct RCC	Collecting ducts; renal medulla	Poor prognosis	Like other RCC	Nests of malignant cells within fibrotic stroma
Wilms Tumor	Metanephric blastema	Pediatric; WAGR, DDS, BWS	Large, soft, tan/grey, well-circumscribed, pseudocapsule	**Triphasic (blastema, stroma, epithelioid)**
Transitional Cell Carcinoma	**Transitional epithelium; renal pelvis & calyces**	**Smoking, chemicals; painless hematuria**	Papillary or flat, multifocal	Pleomorphic nuclei, fibrovascular cores

Practice Question 1

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A 58-year-old male presents to the office complaining of blood in his urine. He denies pain when he urinates, but he does experience left costovertebral pain. He has a 20 pack-year history of smoking, but denies using recreational drugs. After a full workup, a biopsy of his left kidney is taken. Histologic analysis shows round-polygonal cells with clear cytoplasm and delicate branching vasculature. What is the most likely site of origin of this mass?

A. Transitional epithelium
B. Proximal tubule
C. Loop of Henle
D. Distal tubule
E. Collecting duct

Practice Question 1

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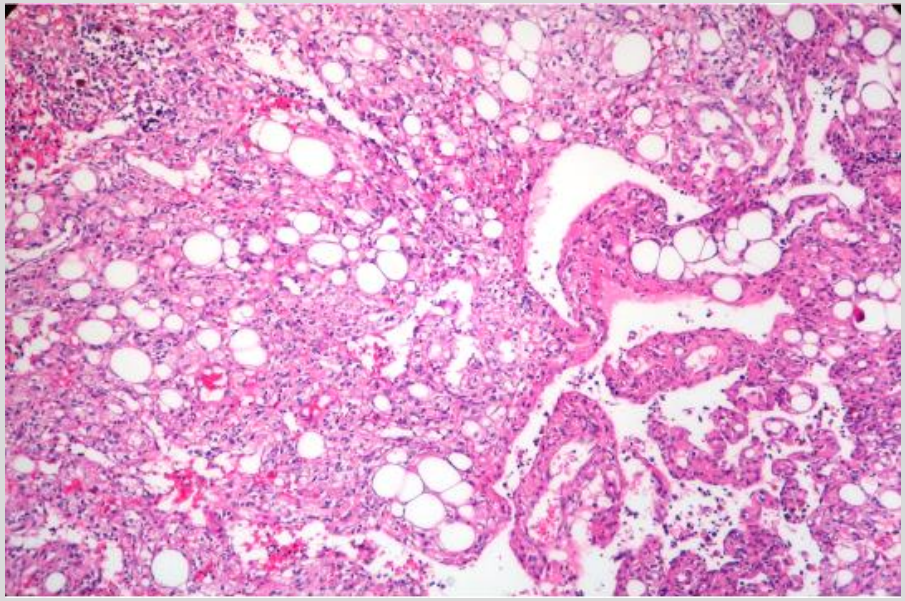
The vignette describes:
Clear cell renal cell carcinoma

Practice Question 2

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A 4-year-old boy presents to the office accompanied by his mother. She explains that she noticed a mass in her son's abdomen that has not gone away with defecation. During the physical exam, areas of cutaneous hypopigmentation are observed on the patient's back as well as multiple erythematous papules on the patient's face. An accompanying biopsy of the patient's abdominal mass is shown. What is the most likely gene mutation associated with this patient's condition?

A. MET
B. NF1
C. VHL
D. TSC1
E. WT1



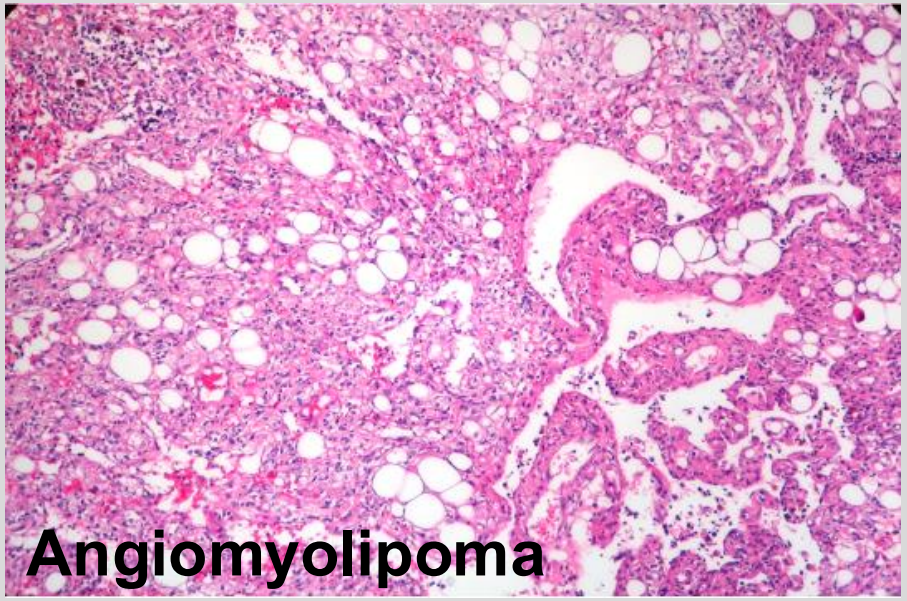
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The vignette describes:
Tuberous Sclerosis



Practice question 3

A 5-year old has complained of abdominal pain for one month. Vital signs are WNL. Physical exam reveals a palpable mass on the right side. Urinalysis reveals hematuria and proteinuria. Abdominal CT shows a 10 cm circumscribed, solid mass in the right kidney. A right-sided nephrectomy is performed and the following gross appearance of the mass is seen in the figure. What is the most likely diagnosis?

A. Neuroblastoma
B. Angiomyolipoma
C. Renal cell carcinoma
D. Wilms tumor
E. Transitional cell carcinoma



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Practice question 3

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A. Neuroblastoma (may cross midline; constitutional symptoms are common)
B. Angiomyolipoma
C. Renal cell carcinoma
D. Wilms tumor
E. Transitional cell carcinoma



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Summary

- Tumors of the kidney range from benign to highly aggressive malignant.
- Papillary adenomas are benign tumors that tend to be smaller than 1.5 cm.
- Angiomyolipoma is associated with Tuberous Sclerosis.
- Oncocytoma is a (mostly) benign epithelial tumor.
- Renal cell carcinoma is the most prevalent kidney tumor in adults and it tends to metastasize early.
- Some types of renal cell carcinoma include: clear cell, papillary, chromophobe, and collecting duct carcinoma.
- The most common pediatric kidney tumor is Wilms tumor (Nephroblastoma), which is associated with syndromes involving *WT1*.

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Lecture Feedback Form

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